Análisis y Desarrollo de Software (ADSO)

GC-F-004v.08

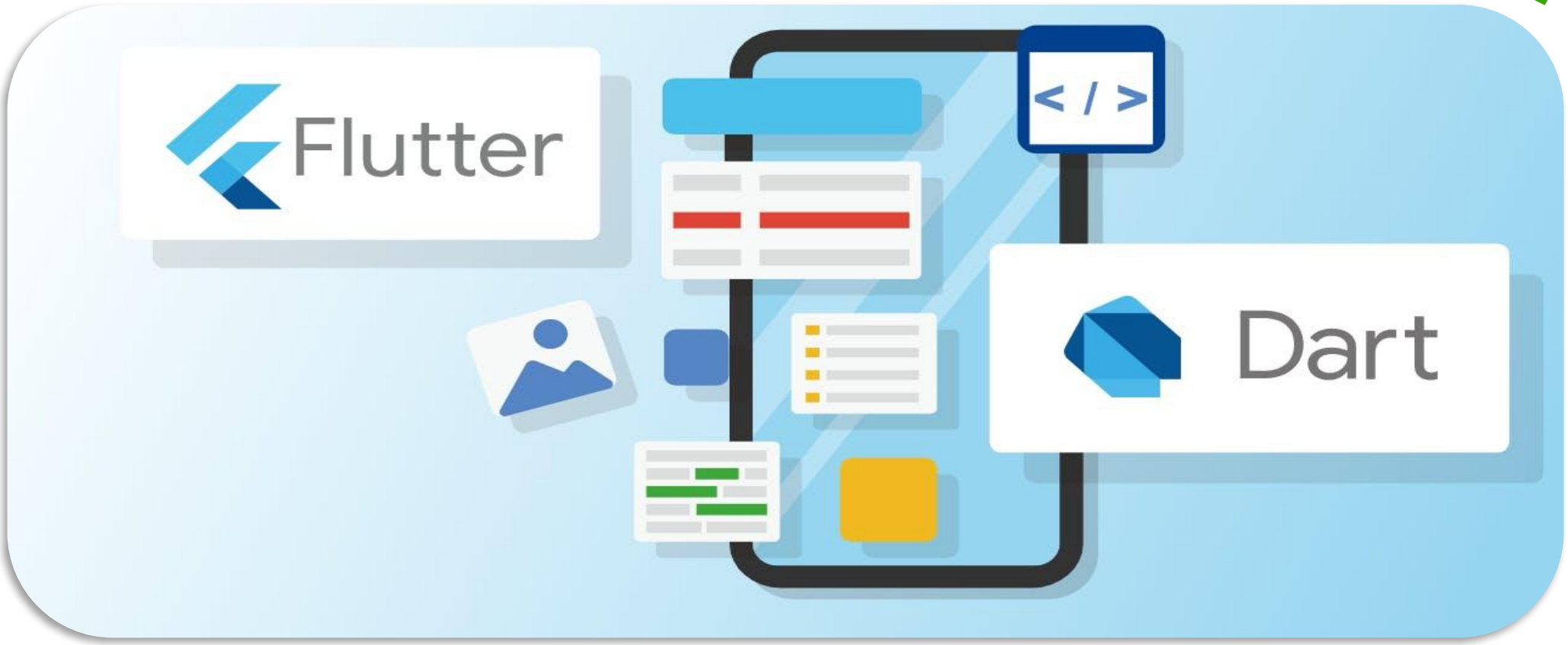
Servicio Nacional de Aprendizaje

TABLA DE **CONTENIDO**

01 FLUTTER - PROYECTOS

FLUTTER

Servicio Nacional de Aprendizaje



FLUTTER - Proyecto

Estructuración de proyecto:

- Determinar las condiciones del proyecto
  - Determinar la arquitectura
  - Determinar las fuentes de datos
- Organizar proyecto (validar elementos compartidos)
  - Variables globales
  - Clases
  - Widget
  - Objetos en común

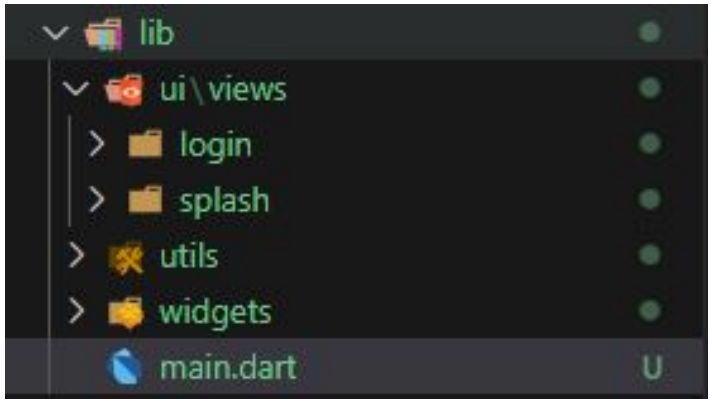
FLUTTER - Proyecto

Crear estructura básica del proyecto

Determinar la estructura del proyecto

- Vistas
- Datos
- Utilidades

Creamos la siguiente estructura de carpetas:



FLUTTER - Proyecto

main.dart => *Clase principal, en esta clase se ejecuta la vista principal, regla de oro (no tocarás el main)*

- Eliminamos la barra de navegación.
- Adicionamos la vista SplashView.

lib > main.dart > ...

```
1  import 'package:flutter/material.dart';
2  import 'package:get/get.dart';
3  import 'package:project_one/ui/views/splash/splash_view.dart';
4
   Run | Debug | Profile
5  void main() {
6    runApp(const MainApp());
7  }
8
9  class MainApp extends StatelessWidget {
10   const MainApp({super.key});
11
12   @override
13   Widget build(BuildContext context) {
14     return const GetMaterialApp(
15       debugShowCheckedModeBanner: false,
16       home: SplashView(),
17     ); // GetMaterialApp
18   }
19 }
20
```

FLUTTER - Proyecto

ui/views/splash/splash_view.dart => *Clase que renderiza el splash*

- Añadimos un color de fondo.
  - Importante adicionar la clase de color.
- Texto centrado; lo pueden cambiar por una imagen.
- Timer: Clase para ejecutar la segunda vista en un tiempo determinado. (Tiempo en segundos)

lib > ui > views > splash > splash_view.dart > SplashView > build

```
1  import 'dart:async';
2  import 'package:get/get.dart';
3  import 'package:flutter/material.dart';
4  import 'package:project_one/ui/views/login/Login.dart';
5  import 'package:project_one/utils/Global.colors.dart';
6
7  class SplashView extends StatelessWidget {
8    const SplashView({super.key});
9
10   @override
11   Widget build(BuildContext context) {
12     Timer(const Duration(seconds: 3), () {
13       Get.to(const LoginView());
14     }); // Timer
15     return Scaffold(
16       backgroundColor: GlobalColors.primaryColorH,
17       body: const Center(
18         child: Text(
19           'Logo',
20           style: TextStyle(
21             color: Colors.white, fontSize: 30, fontWeight: FontWeight.bold), // TextStyle
22         ), // Text
23       ), // Center
24     ); // Scaffold
25   }
26 }
```

FLUTTER - Proyecto

utils/Global.colors.dart => *Clase general de colores*

- Variables con los colores de la aplicación

lib > utils > Global.colors.dart > GlobalColors

```dart
import 'dart:ui';

import 'package:hexcolor/hexcolor.dart';

class GlobalColors {
  static HexColor primaryColorH = HexColor('#6200ee');
  static HexColor secondaryColorH = HexColor('#03DAC6');
  static HexColor secondaryBtnColorH = HexColor('#018786');
  static HexColor successColorH = HexColor('#2dd55b');
  static HexColor dangerColorH = HexColor('#c5000f');
  static HexColor warningColorH = HexColor('#ffc409');
  static HexColor infoColorH = HexColor('#6030ff');
  static HexColor lightColorH = HexColor('#f7f9ff');
  static HexColor mediumColorH = HexColor('#5f5f5f');
  static HexColor darkColorH = HexColor('#2f2f2f');

  static Color primaryColor = const Color(0xff6200ee);
  static Color secondaryColor = const Color(0xff03dac6);
  static Color secondaryBtnColor = const Color(0xff018786);
  static Color successColor = const Color(0xff2dd55b);
  static Color dangerColor = const Color(0xffc5000f);
  static Color infoColor = const Color(0xffffc409);
  static Color lightColor = const Color(0xfff7f9ff);
  static Color mediumColor = const Color(0xff5f5f5f);
  static Color darkColor = const Color(0xff2f2f2f);
}
```

FLUTTER - Proyecto

widgets/Header.dart => *Widget de la cabecera usada en el login, se puede tener varias cabeceras.*

lib > widgets > Header.dart > HeaderLogin > build

```dart
// ignore_for_file: file_names

import 'package:flutter/material.dart';
import 'package:project_one/utils/Global.colors.dart';

class HeaderLogin extends StatelessWidget {
  const HeaderLogin({super.key});

  @override
  Widget build(BuildContext context) {
    // todo: implement build
    return SizedBox(
      height: 250,
      width: double.infinity,
      child: CustomPaint(
        painter: HederLoginPainter(),
      ), // CustomPaint
    ); // SizedBox
  }
}
```

```dart
class HederLoginPainter extends CustomPainter {
  @override
  void paint(Canvas canvas, Size size) {
    // todo: implement build
    final paint = Paint();
    paint.color = GlobalColors.primaryColorH;
    paint.style = PaintingStyle.fill;
    final path = Path();
    path.lineTo(0, size.height * 1.0);
    path.lineTo(
      size.width * 0.2,
      size.height * 1.0,
    );
    path.lineTo(size.width, size.height * 0.8);
    path.lineTo(size.width, 0);
    canvas.drawPath(path, paint);
  }

  @override
  bool shouldRepaint(covariant CustomPainter oldDelegate) {
    // todo: implement build
    return true;
  }
}
```

FLUTTER - Proyecto

widgets/Logo.dart => *Widget de logo*

lib > widgets > Logo.dart > ...

```dart
// ignore_for_file: file_names
import 'package:flutter/material.dart';

class LogoHeader extends StatelessWidget {
  const LogoHeader({super.key});

  @override
  Widget build(BuildContext context) {
    // todo: implement build
    return Positioned(
      top: 80,
      left: MediaQuery.of(context).size.width * 0.38,
      child: Container(
        height: 80,
        width: 80,
        decoration: BoxDecoration(
            color: Colors.white,
            borderRadius: BorderRadius.circular(50),
            boxShadow: const [
              BoxShadow(blurRadius: 10, color: Colors.black26)
            ]), // BoxDecoration
        child: const Align(
          alignment: Alignment.center,
          child: Text(
            'FRAVE',    "FRAVE": Unknown word.
            style: TextStyle(fontSize: 21, fontWeight: FontWeight.bold),
          ), // Text
        ), // Align
      ), // Container
    ); // Positioned
  }
}
```

FLUTTER - Proyecto

widgets/TextFieldCustom.dart => *Widget para las cajas de texto, con este elemento podemos asignar un estilo general a todos nuestros elementos*

lib > widgets > TextFieldCustom.dart > TextFieldCustom > TextFieldCustom

```dart
// ignore_for_file: file_names

import 'package:flutter/material.dart';

import '../utils/Global.colors.dart';

class TextFieldCustom extends StatelessWidget {
  final TextInputType type;
  final IconData icon;
  final Color appColor = GlobalColors.secondaryBtnColorH;
  final bool pass;
  final String label;
  final String hint;
  final double radius = 50;
  TextFieldCustom(
      {required this.type,
      required this.icon,
      super.key,
      required this.label,
      required this.hint,
      this.pass = false});
  @override
```

```dart
  @override
  Widget build(BuildContext context) {
    // todo: implement build
    return TextField(
      keyboardType: type,
      obscureText: pass,
      decoration: InputDecoration(
        prefixIcon: Icon(
          icon,
          color: GlobalColors.mediumColorH,
        ), // Icon
        labelText: label,
        hintText: hint,
        filled: true,
        fillColor: GlobalColors.lightColor,
        border: OutlineInputBorder(
            borderSide: BorderSide(color: GlobalColors.lightColorH),
            borderRadius: BorderRadius.all(Radius.circular(radius))), // OutlineInputBorder
        enabledBorder: OutlineInputBorder(
          borderSide: BorderSide(color: GlobalColors.mediumColor),
          borderRadius: BorderRadius.all(Radius.circular(radius)),
        ), // OutlineInputBorder
      ), // InputDecoration
    ); // TextField
  }
}
```

FLUTTER - Proyecto

ui/views/login/Login.dart => Vista del login

lib > ui > views > login > Login.dart > LoginView > build

```dart
// ignore_for_file: file_names

import 'package:flutter/material.dart';
import 'package:get/get.dart';
import 'package:project_one/ui/views/login/SignUp.dart';
import 'package:project_one/widgets/Header.dart';
import 'package:project_one/widgets/Logo.dart';
import 'package:project_one/widgets/TextFieldCustom.dart';
import '../../../utils/Global.colors.dart';

class LoginView extends StatelessWidget {
  const LoginView({super.key});
  static String id = "login_view";
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: ListView(
        padding: const EdgeInsets.only(top: 0),
        physics: const BouncingScrollPhysics(),
        children: const [
          Stack(
            children: [
              HeaderLogin(),
              LogoHeader(),
            ],
          ), // Stack
          Title(),
          SizedBox(
            height: 40,
          ), // SizedBox
          EmailAndPassword(),
          ForgotPassword(),
          SizedBox(
            height: 40,
          ), // SizedBox
          ButtonSinIn(),
        ],
      ), // ListView
    ); // Scaffold
  }
}
```

```dart
class Title extends StatelessWidget {
  const Title({super.key});

  @override
  Widget build(BuildContext context) {
    // todo: implement build

    return Padding(
      padding: const EdgeInsets.all(15.0),
      child: Row(
        children: [
          const Text(
            'SIGN IN',
            style: TextStyle(fontSize: 25, fontWeight: FontWeight.bold),
          ), // Text
          const Text(
            '/',
            style: TextStyle(fontSize: 25, color: Colors.grey),
          ), // Text
          TextButton(
              onPressed: () {
                Get.to(const SignUp());
              },
              child: const Text(
                'SIGN UP',
                style: TextStyle(
                    fontSize: 18,
                    fontWeight: FontWeight.bold,
                    color: Colors.grey), // TextStyle
              )) // Text // TextButton
        ],
      ), // Row
    ); // Padding
  }
}
```

FLUTTER - Proyecto

ui/views/login/Login.dart => Vista del login

```dart
class EmailAndPassword extends StatelessWidget {
  const EmailAndPassword({super.key});

  @override
  Widget build(BuildContext context) {
    // todo: implement build
    return Padding(
      padding: const EdgeInsets.symmetric(horizontal: 20.0),
      child: Column(
        children: [
          TextFieldCustom(
              type: TextInputType.emailAddress,
              icon: Icons.email_outlined,
              label: 'Email',
              hint: 'Enter email address'), // TextFieldCustom
          const SizedBox(
            height: 40,
          ), // SizedBox
          TextFieldCustom(
              type: TextInputType.visiblePassword,
              icon: Icons.password_outlined,
              label: 'Password',
              hint: 'Enter password',
              pass: true), // TextFieldCustom
        ],
      ), // Column
    ); // Padding
  }
}
```

```dart
class ForgotPassword extends StatelessWidget {
  const ForgotPassword({super.key});

  @override
  Widget build(BuildContext context) {
    // todo: implement build
    return Container(
      padding: const EdgeInsets.only(right: 25, top: 25),
      alignment: Alignment.centerRight,
      child: const Text('Forgot Password'),
    ); // Container
  }
}
```

```dart
class ButtonSinIn extends StatelessWidget {
  const ButtonSinIn({super.key});

  @override
  Widget build(Object context) {
    // todo: implement build
    return Container(
        margin: const EdgeInsets.all(25),
        decoration: BoxDecoration(
            color: GlobalColors.secondaryColorH,
            borderRadius: const BorderRadius.all(Radius.circular(50))), // BoxDecoration
        child: TextButton(
            onPressed: () => {},
            child: const Text(
              'SING IN',
              style: TextStyle(
                  height: 3.1,
                  fontSize: 18,
                  fontWeight: FontWeight.bold,
                  color: Colors.white), // TextStyle
            )); // Text // TextButton // Container
  }
}
```

FLUTTER - Proyecto

ui/views/login/SignUp.dart => Vista del registro

lib > ui > views > login > SignUp.dart > TextFields > build

```dart
// ignore_for_file: file_names

import 'package:flutter/material.dart';
import 'package:project_one/ui/views/login/Login.dart';
import 'package:project_one/widgets/Header.dart';
import 'package:project_one/widgets/Logo.dart';

import '../../../utils/global.colors.dart';
import '../../../widgets/TextFieldCustom.dart';

class SignUp extends StatelessWidget {
  const SignUp({super.key});

  @override
  Widget build(BuildContext context) {
    // todo: implement build

    return Scaffold(
      body: ListView(
        padding: const EdgeInsets.only(top: 0),
        physics: const BouncingScrollPhysics(),
        children: const [
          Stack(
            children: [HeaderLogin(), LogoHeader()],
          ), // Stack
          Title(),
          TextFields(),
          ButtonSIGNUp(),
        ],
      ), // ListView
    ); // Scaffold
  }
}
```

```dart
class Title extends StatelessWidget {
  const Title({super.key});

  @override
  Widget build(BuildContext context) {
    // todo: implement build

    return Padding(
      padding: const EdgeInsets.all(15.0),
      child: Row(
        children: [
          TextButton(
              onPressed: () {
                Navigator.push(context,
                    MaterialPageRoute(builder: (context) => const LoginView()));
              },
              child: const Text(
                'SIGN IN',
                style: TextStyle(
                    fontSize: 18,
                    fontWeight: FontWeight.bold,
                    color: Colors.grey), // TextStyle
              )), // Text // TextButton
          const Text(
            '/',
            style: TextStyle(fontSize: 25, color: Colors.grey),
          ), // Text
          TextButton(
              onPressed: () {
                // Navigator.push(context, MaterialPageRoute(builder: (context) => const SIGNUp()));
              },
              child: Text(
                'SIGN UP',
                style: TextStyle(
                    fontSize: 25,
                    fontWeight: FontWeight.bold,
                    color: GlobalColors.darkColorH), // TextStyle
              )) // Text // TextButton
        ],
      ), // Row
    ); // Padding
  }
}
```

FLUTTER - Proyecto

ui/views/login/SignUp.dart => Vista del registro

lib > ui > views > login > SignUp.dart > TextFields > build

```dart
class TextFields extends StatelessWidget {
  const TextFields({super.key});

  @override
  Widget build(BuildContext context) {
    // todo: implement build
    return Padding(
      padding: const EdgeInsets.symmetric(horizontal: 20.0),
      child: Column(
        children: [
          TextFieldCustom(
              type: TextInputType.text,
              icon: Icons.person_outlined,
              label: 'User',
              hint: 'Enter user'), // TextFieldCustom
          const SizedBox(
            height: 20,
          ), // SizedBox
          TextFieldCustom(
              type: TextInputType.emailAddress,
              icon: Icons.email_outlined,
              label: 'Email',
              hint: 'Enter email address'), // TextFieldCustom
          const SizedBox(
            height: 20,
          ), // SizedBox
          TextFieldCustom(
              type: TextInputType.phone,
              icon: Icons.phone_outlined,
              label: 'Phone',
              hint: 'Enter phone '), // TextFieldCustom
          const SizedBox(
            height: 20,
          ), // SizedBox
          TextFieldCustom(
              type: TextInputType.visiblePassword,
              icon: Icons.password_outlined,
              label: 'Password',
              hint: 'Enter Password',
              pass: true), // TextFieldCustom
          const SizedBox(
            height: 20,
          ), // SizedBox
        ],
      ), // Column
    ); // Padding
  }
}
```

```dart
class ButtonSIGNUp extends StatelessWidget {
  const ButtonSIGNUp({super.key});

  @override
  Widget build(Object context) {
    // todo: implement build
    return Container(
        margin: const EdgeInsets.all(25),
        decoration: BoxDecoration(
            color: GlobalColors.secondaryColorH,
            borderRadius: const BorderRadius.all(Radius.circular(50))), // BoxDecoration
        child: TextButton(
            onPressed: () => {},
            child: const Text(
              'SIGN UP',
              style: TextStyle(
                  height: 3.1,
                  fontSize: 18,
                  fontWeight: FontWeight.bold,
                  color: Colors.white), // TextStyle
            )); // Text // TextButton // Container
  }
}
```

FLUTTER - Proyecto

pubspec.yaml => *Adicionar dependencias*

pubspec.yaml

```yaml
name: project_one
description: "A new Flutter project."
publish_to: 'none'
version: 0.1.0

environment:
  sdk: '>=3.3.4 <4.0.0'

dependencies:
  flutter:
    sdk: flutter
  hexcolor: ^3.0.1
  get: ^4.6.6
  flutter_svg: ^2.0.10+1

dev_dependencies:
  flutter_test:
    sdk: flutter
  flutter_lints: ^3.0.0
  flutter_launcher_icons: ^0.11.0

# flutter_icons:
#   android: true
#   adaptive_icon_background: "#32a852"
#   image_path: "assets/img/icons/icon.png"


flutter:
  uses-material-design: true
```

G R A C I A S

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